

Assignment 2-Case Study

Stage 4 Lab Activities

Implementing the SmartCare Domain Layer

DESIGN FIRST -> AI PAIR PROGRAMMING -> REVIEW -> VERIFY | 1hour

A - Revisit Approved UML

Confirm responsibilities, attributes and relationships before coding.

B - Implement Patient: AI OFF

Implement Patient with type hints and basic validation.

C - Implement Practitioner: AI OFF

Implement Practitioner with identifier, name and specialty; no database logic.

D - Implement Appointment: AI ON

Give AI the approved Appointment UML, business rules and explicit constraints. Ask it to implement only Appointment and agreed enum/exception.

E - Review Generated Code

Check model consistency, unsupported features, public state mutation, unnecessary inheritance, invented dependencies and error handling.

F - Manual Behaviour Checks

Create valid objects, test invalid input, cancel a scheduled appointment and attempt an illegal repeated transition.

G - Refactor

Remove unnecessary code and make implementation simpler and design-consistent.

H - AI Engineering Log

Record prompt, generated contribution, decisions and verification evidence.

Suggested AI prompt

Act as a Python pair programmer. Implement only the Appointment class from the approved SmartCare UML. Use type hints and an AppointmentStatus enum. Cancelled appointments remain as objects. Do not add database, UI, notification or service classes. Protect status transitions and explain any decision not directly visible in the UML.

Reflection

Which AI-generated part did you modify or reject? Why? How did the approved design constrain the AI?